

Carmen Yuchen Guo

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EDUCATION

University of California, San Diego

M.S., Computational Social Science, 4.0/4.0 GPA

La Jolla, CA

Expected June 2026

Relevant Courses: Advanced Statistical Applications, Text as Data, Neural Networks, Causal Inference

The University of Hong Kong

M.S., Marketing

Hong Kong SAR

Graduated July 2022

Relevant Courses: Applied Marketing Research & Metrics, Big Data Consumer Analytics

Peking University

B.A., Advertising

Beijing, China

Graduated July 2021

Relevant Courses: Calculus, Communication Research Method, Principle of Economics

Honors: May 4th Scholarship of Peking University (2018), Academic Excellence Award (2020)

TECHNICAL SKILLS

Statistical Analysis & Modeling: GLMs, multilevel models, supervised and unsupervised methods, regularized regression (Lasso/Ridge), tree-based models, feature engineering, model evaluation, deep learning (CNN, RNN, autoencoders)

Programming & Tools: Python (PyTorch, scikit-learn, Pandas, NumPy, Matplotlib), R, SQL, agentic coding

NLP & Text Analysis: Text preprocessing, topic modeling (LDA), clustering, supervised text classification

Languages: Mandarin (native/bilingual), English (native/bilingual), Cantonese (native/bilingual)

PROFESSIONAL EXPERIENCE

The University of Hong Kong

Program Manager, Admissions Data Analytics

Hong Kong SAR

Dec 2024 – June 2025

Management Trainee

Sep 2023 – Dec 2024

- Selected as one of 7 candidates from 700+ applicants for HKU elite management trainee program
- Built automated Excel workflows (pivot tables, XLOOKUP, advanced formulas) to clean and analyze ~30,000 application records, producing longitudinal and comparative trends with each weekly data update and shortlisting applicants
- Designed and maintained interactive dashboards with Excel and Power BI integrating multiple data sources for benchmarking, enrollment estimation and diversity metrics monitoring, informing strategic admission decisions
- Applied LDA topic modeling on student queries to identify information needs, informing the design and improvement of an academic advising chatbot, achieving 85%+ goal completion post-launch

Sandpiper

Account Executive

Hong Kong SAR

June 2022 – Feb 2023

- Conducted statistical analysis in R on survey data from 8,087 respondents across 8 countries, applying descriptive statistics and hypothesis testing (T-tests, ANOVA) to identify demographic and socio-economic patterns, with year-over-year comparisons against prior survey waves
- Translated findings into data-driven insights that anchored a 50-page published industry report on digital lifestyle
- Built market-specific data snapshots and visualizations for a consumer expectation survey, enabling tailored distribution of survey findings in different markets
- Compiled monthly Media Monitoring and Analysis Reports, summarizing key news coverages and policy updates related to COVID-19 oral antivirals across the region, sharing trends and opportunities
- Synthesized literature and academic publications to support a healthcare policy white paper

RELEVANT PROJECTS

UC San Diego

La Jolla, CA

Graduate Capstone – Deep Learning for Multilevel Regression and Poststratification

Sep 2025 – Present

- Developing an autoencoder-based deep learning extension of multilevel regression and poststratification (MRP) to estimate local-level policy attitudes from national survey data
- Designed a climate change opinion and policy survey, structured for integration with census data to enable estimation through the deep learning model

Research Project – Textual Analysis of Party Influence in Institutional Writing

Sep 2025 – Dec 2025

- Trained and validated supervised text classification models (Naive Bayes, LASSO, Decision Tree) to measure Chinese Communist Party influence in university documents using hand-labeled ideological indicator
- Assembled a novel text corpus by context-aware segmentation of reports; conducted text preprocessing, model comparison, and error analysis to apply predictions at scale to thousands of snippets

The University of Hong Kong

Hong Kong SAR

Research Project – Emotional Motivator Network and TikTok Marketing

Oct 2021

- Visualized and analyzed emotional motivator co-occurrence networks in TikTok contents using Gephi, discovering patterns relating to brand characteristics
- Performed correlation and Multiple Regression analysis with R to quantify the impact of emotional motivators on viewer engagement